

Tutorial 1 – Plate Bending

Questions

A rectangular simply-supported plate has length a in the x -direction and width b in the y -direction. Assume that the plate has a known flexural stiffness D . The plate is subject to a concentrated force F (i.e. a point load) acting at the location $x=x_0, y=y_0$.

- 1) Consider the general Navier's series expansion of the applied pressure load:

$$p(x,y) = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} B_{jk} \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b}$$

Find the coefficients B_{mn} in the expression above corresponding to the applied point load F .
(see the hint on page 2 of this handout for how to proceed)

[30 marks]

- 2) Find the analytical expression of the transverse displacement $w(x,y)$ for the simply supported plate described above subject to the force F .

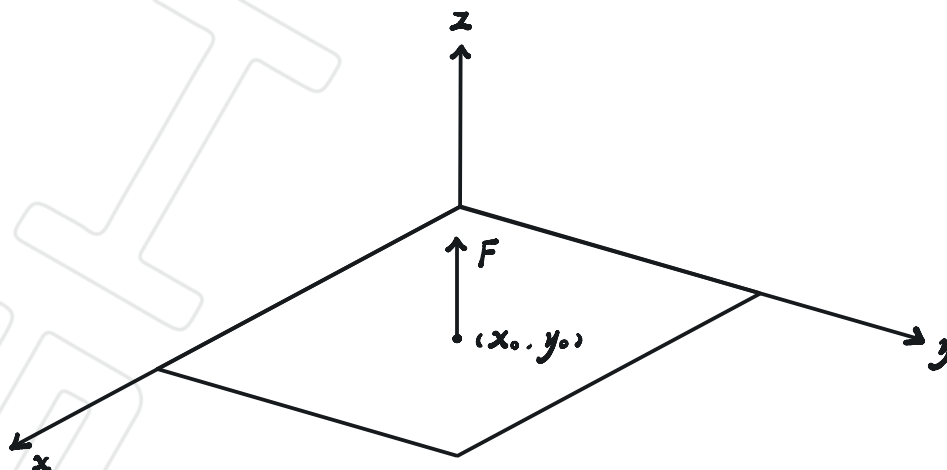
[30 marks]

- 3) Determine the maximum deflection for a square plate ($b=a$) caused by a point load F applied at its centre ($x_0=y_0=a/2$).

[20 marks]

- 4) Find the expression of the moments at the centre of the aforementioned square plate.

[20 marks]



$$1) p(x, y) = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} B_{jk} \sin\left(\frac{j\pi x}{a}\right) \sin\left(\frac{k\pi y}{b}\right)$$

$$\therefore \int_0^a \int_0^b p(x, y) \sin\left(\frac{j'\pi x}{a}\right) \sin\left(\frac{k'\pi y}{b}\right) dx dy$$

$$= \int_0^a \int_0^b \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} B_{jk} \sin\left(\frac{j\pi x}{a}\right) \sin\left(\frac{j'\pi x}{a}\right) \sin\left(\frac{k\pi y}{b}\right) \sin\left(\frac{k'\pi y}{b}\right) dx dy$$

$$= B_{jk} \left(\frac{a}{2}\right) \left(\frac{b}{2}\right)$$

$$\therefore B_{jk} = \frac{4}{ab} \int_0^a \int_0^b p(x, y) \sin\left(\frac{j\pi x}{a}\right) \sin\left(\frac{k\pi y}{b}\right) dx dy$$

$$= \frac{4}{ab} \int_0^a \int_0^b F \delta(x-x_0) \delta(y-y_0) \sin\left(\frac{j\pi x}{a}\right) \sin\left(\frac{k\pi y}{b}\right) dx dy$$

$$= \frac{4F}{ab} \sin\left(\frac{j\pi x_0}{a}\right) \sin\left(\frac{k\pi y_0}{b}\right)$$

$$2) \text{ Let } w = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} A_{jk} \sin\left(\frac{j\pi x}{a}\right) \sin\left(\frac{k\pi y}{b}\right)$$

for z-direction force equilibrium,

$$-\frac{Et^3}{12(1-\nu^2)} \left(\frac{\partial^4 w}{\partial x^4} + 2 \frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4} \right) + p = 0$$

$$\therefore \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} A_{jk} \pi^4 \left(\frac{j^4}{a^4} + 2 \frac{j^2 k^2}{a^2 b^2} + \frac{k^4}{b^4} \right) \sin\left(\frac{j\pi x}{a}\right) \sin\left(\frac{k\pi y}{b}\right) = \frac{12p(1-\nu^2)}{Et^3}$$

$$\therefore \int_0^a \int_0^b \text{LHS} \sin\left(\frac{j'\pi x}{a}\right) \sin\left(\frac{k'\pi y}{b}\right) dx dy = \int_0^a \int_0^b \text{RHS} \sin\left(\frac{j'\pi x}{a}\right) \sin\left(\frac{k'\pi y}{b}\right) dx dy$$

According to the answer 1),

$$\left(\frac{a}{2}\right) \left(\frac{b}{2}\right) A_{jk} \pi^4 \left(\frac{j^4}{a^4} + 2 \frac{j^2 k^2}{a^2 b^2} + \frac{k^4}{b^4} \right) = \frac{1}{D} \left(\frac{a}{2}\right) \left(\frac{b}{2}\right) B_{jk}$$

$$\therefore A_{jk} = \frac{4F \sin\left(\frac{j\pi x_0}{a}\right) \cos\left(\frac{k\pi y_0}{b}\right)}{ab D \pi^4 \left(\frac{j^4}{a^4} + \frac{k^4}{b^4} \right)}$$

$$\therefore w = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{4F}{\pi^4 D ab \left[\left(\frac{j}{a}\right)^4 + \left(\frac{k}{b}\right)^4 \right]} \sin\left(\frac{j\pi x}{a}\right) \sin\left(\frac{j\pi x_0}{a}\right) \sin\left(\frac{k\pi y}{b}\right) \sin\left(\frac{k\pi y_0}{b}\right)$$

$$3) \text{ for } a=b, x_0=y_0=\frac{a}{2}$$

$$w = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{4F}{\pi^4 D a^2 \left[\left(\frac{j}{a}\right)^4 + \left(\frac{k}{a}\right)^4 \right]} \sin\left(\frac{j\pi x}{a}\right) \sin\left(\frac{j\pi}{2}\right) \sin\left(\frac{k\pi y}{a}\right) \sin\left(\frac{k\pi}{2}\right)$$

$$\therefore \frac{\partial w}{\partial x} = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{4F}{\pi^4 D a^2 \left[\left(\frac{j}{a}\right)^4 + \left(\frac{k}{a}\right)^4 \right]} \left(\frac{j\pi}{a}\right) \cos\left(\frac{j\pi x}{a}\right) \sin\left(\frac{j\pi}{2}\right) \sin\left(\frac{k\pi y}{a}\right) \sin\left(\frac{k\pi}{2}\right) = 0$$

$$\therefore x|_{w_{max}} = \frac{a}{2}$$

similarly, $y|_{w_{max}} = \frac{b}{2}$

$$\therefore w\left(\frac{a}{2}, \frac{b}{2}\right) = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{4F}{\pi^2 D a^2 \left[\left(\frac{j}{a}\right)^2 + \left(\frac{k}{b}\right)^2\right]} \sin^2\left(\frac{j\pi}{2}\right) \sin^2\left(\frac{k\pi}{2}\right)$$

$$= \begin{cases} 0 & \cdot j \text{ or } k \text{ is even} \\ \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{4F}{\pi^2 D \left(\frac{j^2}{a^2} + \frac{k^2}{b^2}\right)} & \cdot j \text{ and } k \text{ are odd} \end{cases}$$

$$\therefore w_{max} = \sum_{j=1,3,5,\dots}^{\infty} \sum_{k=1,3,5,\dots}^{\infty} \frac{4Fa^2}{\pi^2 D (j^2 + k^2)^2}$$

$$4) M_x = -D \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right)$$

$$M_y = -D \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right)$$

$$M_{xy} = -D(1-\nu) \frac{\partial^2 w}{\partial x \partial y}$$

where

$$\frac{\partial w}{\partial y} = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{4Fa^2}{\pi^2 D (j^2 + k^2)^2} \left(\frac{k\pi}{b} \right) \sin\left(\frac{j\pi x}{a}\right) \sin\left(\frac{j\pi}{2}\right) \cos\left(\frac{k\pi y}{b}\right) \sin\left(\frac{k\pi}{2}\right)$$

$$\therefore \frac{\partial^2 w}{\partial x^2} = \left(\frac{j\pi}{a} \right)^2 w$$

$$\frac{\partial^2 w}{\partial y^2} = \left(\frac{k\pi}{b} \right)^2 w$$

$$\frac{\partial^2 w}{\partial x \partial y} = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{4Fa^2}{\pi^2 D (j^2 + k^2)^2} \left(\frac{j\pi}{a} \right) \cos\left(\frac{j\pi x}{a}\right) \sin\left(\frac{j\pi}{2}\right) \cos\left(\frac{k\pi y}{b}\right) \sin\left(\frac{k\pi}{2}\right)$$

$$\therefore \frac{\partial^2 w}{\partial x^2} \Big|_{\left(\frac{a}{2}, \frac{b}{2}\right)} = \left(\frac{j\pi}{a} \right)^2 \sum_{j=1,3,5,\dots}^{\infty} \sum_{k=1,3,5,\dots}^{\infty} \frac{4Fa^2}{\pi^2 D (j^2 + k^2)^2}$$

$$\frac{\partial^2 w}{\partial y^2} \Big|_{\left(\frac{a}{2}, \frac{b}{2}\right)} = \left(\frac{k\pi}{b} \right)^2 \sum_{j=1,3,5,\dots}^{\infty} \sum_{k=1,3,5,\dots}^{\infty} \frac{4Fa^2}{\pi^2 D (j^2 + k^2)^2}$$

$$\frac{\partial^2 w}{\partial x \partial y} \Big|_{\left(\frac{a}{2}, \frac{b}{2}\right)} = 0$$

$$\therefore M_x = -D \left[\left(\frac{j\pi}{a} \right)^2 \sum_{j=1,3,5,\dots}^{\infty} \sum_{k=1,3,5,\dots}^{\infty} \frac{4Fa^2}{\pi^2 D (j^2 + k^2)^2} + \nu \left(\frac{k\pi}{b} \right)^2 \sum_{j=1,3,5,\dots}^{\infty} \sum_{k=1,3,5,\dots}^{\infty} \frac{4Fa^2}{\pi^2 D (j^2 + k^2)^2} \right]$$

$$= -D \sum_{j=1,3,5,\dots}^{\infty} \sum_{k=1,3,5,\dots}^{\infty} \frac{4F(j^2 + \nu k^2)}{\pi^2 D (j^2 + k^2)^2}$$

$$M_y = -D \left[\left(\frac{k\pi}{b} \right)^2 \sum_{j=1,3,5,\dots}^{\infty} \sum_{k=1,3,5,\dots}^{\infty} \frac{4Fa^2}{\pi^2 D (j^2 + k^2)^2} + \nu \left(\frac{j\pi}{a} \right)^2 \sum_{j=1,3,5,\dots}^{\infty} \sum_{k=1,3,5,\dots}^{\infty} \frac{4Fa^2}{\pi^2 D (j^2 + k^2)^2} \right]$$

$$= -D \sum_{j=1,3,5,\dots}^{\infty} \sum_{k=1,3,5,\dots}^{\infty} \frac{4F(k^2 + \nu j^2)}{\pi^2 D (j^2 + k^2)^2}$$

$$M_{xy} = 0$$

Hint: The load due to a concentrated F force acting on the plate surface at the point (x_0, y_0) can be expressed via the two-dimensional Dirac's delta function as:

$$p(x, y) = F \delta(x - x_0) \delta(y - y_0)$$

Remember that the Dirac's delta function satisfies the properties:

$$P1) \quad \int_0^a \int_0^b \delta(x - x_0) \delta(y - y_0) dx dy = 1;$$

$$P2) \quad \int_0^a \int_0^b f(x, y) \delta(x - x_0) \delta(y - y_0) dx dy = f(x_0, y_0),$$

where $f(x, y)$ is an arbitrary continuous function defined in the domain $[0, a] \times [0, b]$. Hence, we have this useful result:

$$\int_0^a \int_0^b \delta(x - x_0) \delta(y - y_0) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy = \sin \frac{m\pi x_0}{a} \sin \frac{n\pi y_0}{b}.$$

$$p(x, y) = \sum \sum p_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

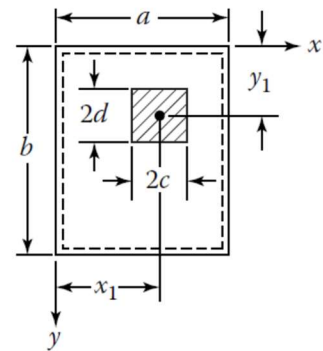
$$F = \int_A p(x, y) dx dy \Rightarrow \text{let } p(x, y) = \frac{F}{4cd} \Rightarrow$$

$$F = F = \int_{x_1-c}^{x_1+c} \int_{y_1-d}^{y_1+d} \frac{F}{4cd} dx dy = F \int_{x_1-c}^{x_1+c} \int_{y_1-d}^{y_1+d} \frac{1}{4cd} dx dy$$

let $c, d = 0$ but keep the area under integral constant as $= 1 \Rightarrow$

$$\text{point load} \Rightarrow F = \int_{x_1}^{x_1} \int_{y_1}^{y_1} F \delta(x - x_1) \delta(y - y_1) dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F \delta(x - x_1) \delta(y - y_1) dx dy$$

$$p(x, y) = F \delta(x - x_1) \delta(y - y_1)$$



OR, Alternative Solution

$$p_{mn} = \frac{P}{abcd} \int_{y_1-d}^{y_1+d} \int_{x_1-c}^{x_1+c} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy$$

$$p_{mn} = \frac{4P}{\pi^2 mncd} \sin \frac{m\pi x_1}{a} \sin \frac{n\pi y_1}{b} \sin \frac{m\pi c}{a} \sin \frac{n\pi d}{b}.$$

Tutorial 1, Plate Bending

Preliminary Considerations (Revisiting L)

First of all, let us introduce the Kronecker delta, which is defined as follows

$$\delta_{mn} = \begin{cases} 1, & m = n \\ 0, & m \neq n \end{cases}$$

where m and n are positive integers. Let us also recall the following orthogonality property of sine functions

$$\int_0^a \sin \frac{m\pi x}{a} \sin \frac{n\pi x}{a} dx = \frac{a}{2} \delta_{mn}$$

Following Navier's approach, the pressure acting on a simply-supported plate is expanded as a double sine series in x and y , i.e.

$$p(x, y) = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} B_{jk} \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b}$$

In order to find the coefficients B_{jk} we can multiply both terms of the last equation by sine harmonics in x and y ; thus

$$p(x, y) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} B_{jk} \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

Integrating the above with respect to the plate area we have

$$\begin{aligned} \int_0^a \int_0^b p(x, y) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy &= \\ &= \int_0^a \int_0^b \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} B_{jk} \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy \end{aligned}$$

Hence

$$\begin{aligned} \int_0^a \int_0^b p(x, y) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy &= \\ &= \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} B_{jk} \int_0^a \sin \frac{j\pi x}{a} \sin \frac{m\pi x}{a} dx \int_0^b \sin \frac{k\pi y}{b} \sin \frac{n\pi y}{b} dy \end{aligned}$$

Exploiting the orthogonality property of the sine harmonics we obtain

$$\int_0^a \int_0^b p(x, y) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy = \frac{ab}{4} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} B_{jk} \delta_{jm} \delta_{kn} = \frac{ab}{4} B_{mn}$$

Hence, we have the following fundamental result

$$B_{mn} = \frac{4}{ab} \int_0^a \int_0^b p(x, y) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy$$

Similarly, in Navier's approach the transverse displacement of a simply supported plate is expanded as a double sine series

$$w(x, y) = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} w_{jk} \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b}$$

Note that the series above satisfies exactly the boundary conditions of the problem for a simply-supported plate, i.e. zero displacements and zero bending moments (zero curvatures) on all edges.

The exact relation between the w_{jk} and B_{jk} coefficients is found by substituting the sine series into the plate equilibrium equation

$$\frac{\partial^4 w}{\partial x^4} + 2 \frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4} = \frac{p}{D}$$

Substituting the displacement expression into the left hand side of the last equation, we find

$$\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} w_{jk} \left[\left(\frac{j\pi}{a} \right)^2 + \left(\frac{k\pi}{b} \right)^2 \right]^2 \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b} = \frac{p}{D}$$

Multiplying both terms by sine harmonics in x and y and integrating with respect to the plate area, we obtain

$$\begin{aligned} \int_0^a \int_0^b \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} w_{jk} \left[\left(\frac{j\pi}{a} \right)^2 + \left(\frac{k\pi}{b} \right)^2 \right]^2 \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy = \\ = \frac{1}{D} \int_0^a \int_0^b \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} p \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \end{aligned}$$

Exploiting the orthogonality property of sine functions, we have

$$\frac{ab}{4} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} w_{jk} \left[\left(\frac{j\pi}{a} \right)^2 + \left(\frac{k\pi}{b} \right)^2 \right]^2 \delta_{jk} \delta_{jm} = \frac{1}{D} \int_0^a \int_0^b \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} p \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

Therefore

$$w_{mn} \left[\left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \right]^2 = \frac{1}{D} \frac{4}{ab} \int_0^a \int_0^b \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} p \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

Note that on the right hand side we now have the B_{mn} coefficients divided by the flexural stiffness of the plate, hence

$$w_{mn} = \frac{1}{D} \left[\left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \right]^{-2} B_{mn}$$

Tutorial 1, Plate bending

Solutions

Q1) As outlined in the tutorial handout, a concentrated force F acting on the plate can be expressed as an “impulse” in 2D space as

$$p(x, y) = F\delta(x - x_0)\delta(y - y_0)$$

where $\delta(x - x_0)$ and $\delta(y - y_0)$ are the Dirac delta functions with respect to x and y . The B_{jk} coefficients are given by

$$B_{jk} = \frac{4}{ab} \int_0^a \int_0^b p(x, y) \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b} dx dy$$

Substituting the pressure associated with a concentrated force into the last equation we have

$$B_{jk} = \frac{4F}{ab} \int_0^a \int_0^b \delta(x - x_0)\delta(y - y_0) \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b} dx dy$$

Exploiting the properties of the Dirac delta function we obtain

$$B_{jk} = \frac{4F}{ab} \sin \frac{j\pi x_0}{a} \sin \frac{k\pi y_0}{b}$$

Q2) The Navier’s expressions of the displacement field for a simply supported plate is

$$w(x, y) = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} w_{jk} \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b}$$

where the coefficients of the series are expressed as

$$w_{jk} = \frac{1}{D} \left[\left(\frac{j\pi}{a} \right)^2 + \left(\frac{k\pi}{b} \right)^2 \right]^{-2} B_{jk}$$

Substituting the expression of the B_{jk} coefficients into the last equation leads to

$$w_{jk} = \frac{4F}{abD} \left[\left(\frac{j\pi}{a} \right)^2 + \left(\frac{k\pi}{b} \right)^2 \right]^{-2} \sin \frac{j\pi x_0}{a} \sin \frac{k\pi y_0}{b}$$

Hence the displacement field is given by

$$w(x,y) = \frac{4F}{abD} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \left[\left(\frac{j\pi}{a} \right)^2 + \left(\frac{k\pi}{b} \right)^2 \right]^{-2} \sin \frac{j\pi x_0}{a} \sin \frac{k\pi y_0}{b} \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{b}$$

Q3) For a square plate with a point load applied at its centre, we have

$$b = a, x_0 = y_0 = \frac{a}{2} \Rightarrow w(x,y) = \frac{4Fa^2}{\pi^4 D} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j^2 + k^2)^2} \sin \frac{j\pi}{2} \sin \frac{k\pi}{2} \sin \frac{j\pi x}{a} \sin \frac{k\pi y}{a}$$

Note that in the series above all the even terms in j and k are zero.

The maximum displacement is achieved at the centre of the plate, where it is straightforward to check that the derivatives of $w(x,y)$ (i.e. the rotations) are zero; hence

$$w\left(\frac{a}{2}, \frac{a}{2}\right) = \frac{4Fa^2}{\pi^4 D} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j^2 + k^2)^2}$$

where only the odd terms must be considered.

Q4) The second derivatives of the displacement for the square plate are

$$\begin{aligned} \frac{\partial^2 w}{\partial x^2} &= -\frac{4F}{\pi^2 D} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{j^2}{(j^2 + k^2)^2} \sin \frac{j\pi x_0}{a} \sin \frac{j\pi x}{a} \sin \frac{k\pi y_0}{a} \sin \frac{k\pi y}{a} \\ \frac{\partial^2 w}{\partial y^2} &= -\frac{4F}{\pi^2 D} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{k^2}{(j^2 + k^2)^2} \sin \frac{j\pi x_0}{a} \sin \frac{j\pi x}{a} \sin \frac{k\pi y_0}{a} \sin \frac{k\pi y}{a} \\ \frac{\partial^2 w}{\partial x \partial y} &= \frac{4F}{\pi^2 D} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{jk}{(j^2 + k^2)^2} \sin \frac{j\pi x_0}{a} \cos \frac{j\pi x}{a} \sin \frac{k\pi y_0}{a} \cos \frac{k\pi y}{a} \end{aligned}$$

Hence, at the centre of the plate we find

$$\begin{aligned} \frac{\partial^2 w}{\partial x^2} &= -\frac{4F}{\pi^2 D} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{j^2}{(j^2 + k^2)^2} \\ \frac{\partial^2 w}{\partial y^2} &= -\frac{4F}{\pi^2 D} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{k^2}{(j^2 + k^2)^2} \\ \frac{\partial^2 w}{\partial x \partial y} &= 0 \end{aligned}$$

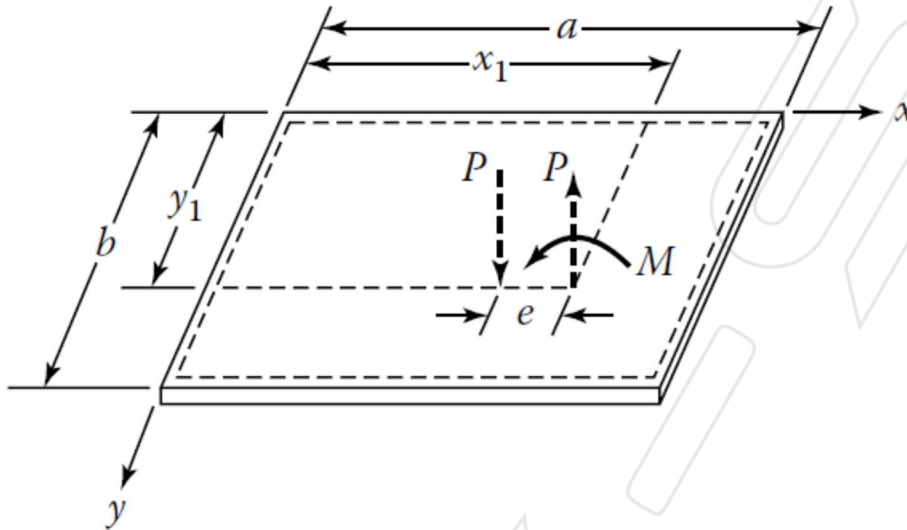
Therefore $M_{xy}(a/2, a/2) = 0$. For the bending moments we find

$$M_x = \frac{4F}{\pi^2} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{j^2 + \nu k^2}{(j^2 + k^2)^2}, \quad M_y = \frac{4F}{\pi^2} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{\nu j^2 + k^2}{(j^2 + k^2)^2}$$

where ν is the Poisson's ratio.

How about this problem:

A simply supported rectangular plate is subjected to a moment M at $x = x_1, y = y_1$. Determine an equation for the deflection.



$$\text{Let } w = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

$$q = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

for force equilibrium in z - direction

$$-D \left(\frac{\partial^4 w}{\partial x^4} + 2 \frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4} \right) + p = 0$$

$$\begin{aligned} \therefore \int_0^a \int_0^b \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \pi^4 \left[\frac{m^4}{a^4} + 2 \frac{m^2 n^2}{a^2 b^2} + \frac{n^4}{b^4} \right] \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \sin \frac{m'\pi x}{a} \sin \frac{n'\pi y}{b} dx dy \\ = \frac{1}{D} \int_0^a \int_0^b \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \sin \frac{m'\pi x}{a} \sin \frac{n'\pi y}{b} dx dy \end{aligned}$$

for orthogonality,

$$\left(\frac{a}{2} \right) \left(\frac{b}{2} \right) A_{mn} \pi^4 \left[\left(\frac{m}{a} \right)^4 + \left(\frac{n}{b} \right)^4 \right] = \frac{1}{D} \left(\frac{a}{2} \right) \left(\frac{b}{2} \right) B_{mn}$$

$$\therefore A_{mn} = \frac{B_{mn}}{D \pi^4 \left[\left(\frac{m}{a} \right)^4 + \left(\frac{n}{b} \right)^4 \right]}$$

where

$$\int_0^a \int_0^b q \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy$$

$$= \int_0^a \int_0^b \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \sin \frac{m'\pi x}{a} \sin \frac{n'\pi y}{b} dx dy$$

for orthogonality,

$$\int_0^a \int_0^b q \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy = \left(\frac{a}{2}\right)\left(\frac{b}{2}\right) B_{mn}$$

$$\therefore B_{mn} = \frac{4}{ab} \int_0^a \int_0^b q \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy$$

decouple M as shown in figure above.

$$M = Pe \text{ for } e \rightarrow 0$$

$$\therefore p = P(x, y) - P(x, -e, y)$$

$$\therefore B_{mn}$$

$$= \frac{4}{ab} \int_0^a \int_0^b [P \delta(x - x_1) \delta(y - y_1) - P \delta(x - x_1 + e) \delta(y - y_1)] \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy$$

$$= \frac{4}{ab} [P \sin \frac{m\pi x_1}{a} \sin \frac{n\pi y_1}{b} - P \sin \frac{m\pi (x_1 - e)}{a} \sin \frac{n\pi y_1}{b}]$$

$$= \frac{4M}{ab} \lim_{e \rightarrow 0} \left[\frac{1}{e} \left(\sin \frac{m\pi x_1}{a} - \sin \frac{m\pi (x_1 - e)}{a} \right) \sin \frac{n\pi y_1}{b} \right]$$

$$\therefore W = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

$$= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{B_{mn}}{D\pi^2 \left[\left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2 \right]} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

$$= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{4M}{D\pi^2 ab \left[\left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2 \right]} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \lim_{e \rightarrow 0} \left[\frac{1}{e} \left(\sin \frac{m\pi x_1}{a} - \sin \frac{m\pi (x_1 - e)}{a} \right) \sin \frac{n\pi y_1}{b} \right]$$

$$= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{4M}{D\pi^2 ab \left[\left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2 \right]} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \frac{d}{dx_1} \left(\sin \frac{m\pi x_1}{a} \sin \frac{n\pi y_1}{b} \right)$$

$$* \frac{df}{dx} = \frac{f(x+h) - f(x)}{h}$$